P DESCENDS ALONG A TAU-COVER PRECOMPOSITION WHEN P IS TAU-LOCAL ON THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 744-758 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 760-767, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 41 of queue_stacks-properties.json (pb=330, stmt 744-758, proof 760-767).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Lemma 0.1. *Let P be a property of morphisms of algebraic spaces as above. Let $\tau \in \{\text{étale}, \text{smooth}, \text{syntomic}, \text{fppf}\}$. Let $\mathcal{X} \to \mathcal{Y}$ and $\mathcal{Y} \to \mathcal{Z}$ be morphisms of algebraic stacks representable by algebraic spaces. Assume*

(1) *$\mathcal{X} \to \mathcal{Y}$ is surjective and étale, smooth, syntomic, or flat and locally of finite presentation,*
(2) *the composition has P, and*
(3) *P is local on the source in the τ topology.*

Then $\mathcal{Y} \to \mathcal{Z}$ has property P.

Proof. Let Z be a scheme and let $Z \to \mathcal{Z}$ be a morphism. Set $X = \mathcal{X} \times_{\mathcal{Z}} Z$, $Y = \mathcal{Y} \times_{\mathcal{Z}} Z$. By (1) $\{X \to Y\}$ is a τ covering of algebraic spaces and by (2) $X \to Z$ has property P. By (3) this implies that $Y \to Z$ has property P and we win. □

Date: stacks-project:stacks-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.